Error-Sensitive Alignment Bounds for Affine-Line Decoding: A Lean-Verified Refinement

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Abstract

Goyal and Guruswami derive mutual correlated agreement (MCA) for the affine-line generator from line decodability, and ask for $n+1$ aligned challenges, where n is the block length. We show that $\max(\lfloor \delta n \rfloor + 2,\ 3)$ aligned challenges suffice when $n < |\mathbb{F}|$, where δ is the proximity parameter. The proof replaces an injection of challenges into coordinates by a double count of mismatch coordinates, using that every close challenge misses at most $\lfloor \delta n \rfloor$ coordinates. The same change in the boosting step lowers the main term of the capacity-regime MCA bound for folded Reed–Solomon codes from nt to $(\delta n + 2)t$: $\varepsilon_{\mathrm{mca}}(C, \delta) \le ((\delta n + 2)t + 3t^3)/|\mathbb{F}|$ instead of $(nt + 3t^3)/|\mathbb{F}|$, with $\delta = 1 - \rho - 2/t$. The gain is a constant factor of about $1/\delta$ on the main term. All statements are proved in Lean 4 on top of ArkLib, without `sorry` and with no axioms beyond the standard ones.

1 Introduction

Proximity gaps and correlated agreement are the soundness engine of FRI-type protocols. Mutual correlated agreement (MCA) asks that, for all but few challenges γ, if $u_0 + \gamma u_1$ agrees with a codeword on a set S of at least $(1-\delta)n$ coordinates, then u_0 and u_1 both agree with codewords on the same S.

Goyal and Guruswami [1] introduce *line decodability* (Definition 3.1): whenever many challenges have a decoded codeword close to the line, a positive fraction of them lie on one affine pencil $c_0 + \gamma c_1$. Their Theorem 3.5 turns $(\ell, \delta, a, \ell n + 1)$ curve decodability into MCA error a/q; for lines this asks for $n+1$ aligned challenges. Since line decodability is usually established for a small number of aligned challenges, it is first boosted using a list-size bound (their Theorem 3.6 and Corollary 3.9), and the boost to $n+1$ costs a term nL, giving error $(nL + a)/q$. The same n appears in the folded Reed–Solomon (FRS) bound [1, Corollary 4.10], $\varepsilon_{\mathrm{mca}} \le (nt + 3t^3)/q$, and in later work [2].

Contribution. The threshold $n+1$ comes from an injection: each aligned challenge produces a coordinate where the line agrees with the pencil but (u_0, u_1) does not, and distinct challenges produce distinct coordinates. We observe that each such coordinate also lies in the agreement set of its challenge, which misses at most $e = \lfloor \delta n \rfloor$ coordinates. A double count then bounds the aligned challenges by $e + 1$ (Lemma 3). Consequences:

- MCA error a/q from (δ, a, b) line decodability for every $b \ge \max(e + 2, 3)$, when $n < q$ (Theorem 4).

- The boost can aim at $e+2$ instead of $n+1$, at cost $(e+1)L$ instead of nL (Lemma 5).
- For FRS codes, $\varepsilon_{\mathrm{mca}}(C,\delta) \le ((\delta n+2)t+3t^3)/q$ (Corollary 7).

The counting lemma itself is not new: the same argument appears as `exists_common_affine_set` in the 68.15-bit Proximity Prize submission (author handle BitWonka, Apache-2.0) [5], where it serves an alignment bound for Reed–Solomon codes. We did not find the error-dependent threshold for line decoding, or the resulting FRS bound, in [1, 3, 2]; the argument is elementary and may be known.

All results are formalized in Lean 4 against ArkLib [6] (Section 6).

2 Setting

Let ι be a finite set of $n \ge 1$ coordinates, $\mathbb{F}$ a finite field with q elements, A an $\mathbb{F}$-module (the alphabet; $A=\mathbb{F}$ for Reed–Solomon codes and $A=\mathbb{F}^s$ for folded codes), and $C \subseteq A^\iota$ an $\mathbb{F}$-linear code. Fix $\delta \in [0,1)$ and put $e = \lfloor \delta n \rfloor$. A word x is *δ-close* to y if they agree on at least $n-e$ coordinates. For $S \subseteq \iota$ write $C|_S$ for the restriction of C to S.

Definition 1 (bad challenge)**.** For $u=(u_0,u_1) \in (A^\iota)^2$, a challenge $\gamma \in \mathbb{F}$ is *bad* if there is $S \subseteq \iota$ with $|S| \ge (1-\delta)n$ such that $(u_0+\gamma u_1)|_S \in C|_S$ but $u_0|_S \notin C|_S$ or $u_1|_S \notin C|_S$. The MCA error $\varepsilon_{\mathrm{mca}}(C,\delta)$ is the maximum over u of $\Pr_\gamma[\gamma \text{ bad}]$, with γ uniform in $\mathbb{F}$.

Definition 2 (line decodability [1, Def. 3.1])**.** C is (δ,a,b) line decodable if for all $f_0,f_1 \in A^\iota$ and all $V : \mathbb{F} \to C$ with $\Pr_\gamma[V(\gamma)\ \delta\text{-close to } f_0+\gamma f_1] \ge a/q$ there are $c_0,c_1 \in C$ with $\Pr_\gamma[V(\gamma)\ \delta\text{-close to } f_0+\gamma f_1 \text{ and } V(\gamma)=c_0+\gamma c_1] \ge b/q$.

3 The counting lemma

Lemma 3. *Let $u_0,u_1,c_0,c_1 \in A^\iota$, $E \subseteq \mathbb{F}$ and $e \in \mathbb{N}$. Suppose that for every $\gamma \in E$ there is $G_\gamma \subseteq \iota$ with*

(A1) $|G_\gamma| \ge n-e$;

(A2) $u_0(i)+\gamma u_1(i) = c_0(i)+\gamma c_1(i)$ for all $i \in G_\gamma$;

(A3) some $i \in G_\gamma$ has $(u_0(i),u_1(i)) \ne (c_0(i),c_1(i))$.

Then $|E| \le e+1$.

Proof. Let $M = \{i : (u_0(i),u_1(i)) \ne (c_0(i),c_1(i))\}$ and $d_k = u_k - c_k$. If $i \in G_\gamma \cap G_\beta \cap M$ with $\gamma \ne \beta$, then by (A2) $(\gamma-\beta)\,d_1(i)=0$, so $d_1(i)=0$ and then $d_0(i)=0$, contradicting $i \in M$. Hence the sets $M \cap G_\gamma$ are pairwise disjoint, and by (A3) nonempty, so $|M| \ge |E|$. Count pairs (i,γ) with $i \in M$, $\gamma \in E$, $i \notin G_\gamma$: by (A1) there are at most $|E|e$; each $i \in M$ lies in at most one G_γ, so there are at least $|M|(|E|-1) \ge |E|(|E|-1)$. Thus $|E|-1 \le e$. □

Only the field structure (to cancel $\gamma-\beta$) and (A1)–(A3) are used; c_0,c_1 need not be codewords. The bound is attained whenever $n \ge e+1$ and $q \ge e+1$: take challenges $z_1,\dots,z_{e+1}$, coordinates $x_1,\dots,x_{e+1}$, $d_1(x_j)=1$, $d_0(x_j)=-z_j$, $d=0$ elsewhere, and $G_{z_j} = \iota \setminus \{x_k : k \ne j\}$.

4 MCA from line decodability

Theorem 4. *Let $\delta < 1$, $b \geq \lfloor \delta n \rfloor + 2$, $b \geq 3$ and $n < q$. If C is (δ, a, b) line decodable, then $\varepsilon_{\mathrm{mca}}(C, \delta) \leq a/q$.*

Proof outline. We follow the proof of [1, Thm. 3.5] as formalized in ArkLib and change one step. Fix u, let B be the set of bad challenges and suppose $|B| > a$. For $\gamma \in B$ fix a witness set S_γ and codeword w_γ.

Nonzero code. Let J be the set of coordinates where some codeword is nonzero; $|J| \leq n < q$ gives a codeword d nonzero on all of J. The set $Z = \{i \notin J : u_0(i) = u_1(i) = 0\}$ has fewer than $n - e$ elements: otherwise $V(\gamma) = \gamma^2 d$ is close to the line for every γ, and line decodability would give at least $b \geq 3$ challenges with $\gamma^2 d = c_0 + \gamma c_1$, which happens for at most two. Define $V(\gamma) = w_\gamma$ on B; off B, if every codeword is close to the line, $V(\gamma) = t_\gamma d$ with t_γ chosen so that $V(\gamma)$ differs from the line on all of J (possible since $|J| < q$), and otherwise a codeword far from the line. V depends only on u and the chosen witnesses, and line decodability holds for every V. Since $|B| > a$, it yields c_0, c_1 and a set E of at least b challenges where $V(\gamma)$ is close to the line and equals $c_0 + \gamma c_1$. Every $\gamma \in E$ has a coordinate where the line equals the pencil but $u \neq (c_0, c_1)$: for $\gamma \in B$ inside S_γ, by the failure of the witness; for $\gamma \notin B$ in the agreement set of $t_\gamma d$ outside Z and J. This is where [1] uses an injection into ι to get $|E| \leq n$. Instead apply Lemma 3 with $G_\gamma = \{i : u_0(i) + \gamma u_1(i) = c_0(i) + \gamma c_1(i)\}$, which has at least $n - e$ elements because $V(\gamma) = c_0 + \gamma c_1$ is close to the line. So $|E| \leq e + 1 < b$, a contradiction.

Zero code. Lemma 3 with pencil $(0, 0)$ and $G_\gamma = S_\gamma$ gives $|B| \leq e + 1$; the synthetic construction of [1] produces $E \subseteq B$ with $|E| \geq b$.

All probabilities are exact counts over a uniform γ; no union bound is used. □

For $n \geq 2$ the hypotheses follow from $b = n + 1$, so Theorem 4 contains [1, Thm. 3.5] for lines. The condition $n < q$ replaces $n + 1 \leq q$, which $b = n + 1$ implies; it excludes Reed–Solomon codes on the whole field. Whether $b = e + 1$ suffices is open; the counting lemma is tight, but that does not settle the implication.

5 Boosting and folded Reed–Solomon codes

Lemma 5. *Under the hypotheses of the boosting lemma of [1] as formalized in ArkLib (`strong_LineDecodable_boost_of_lambda_le`: strong line decodability with parameters (a, b), $b > 1$, list size at most L at radius $\delta b/(b-1)$, $(L+1)^2 < q$), the code is strongly line decodable with parameters $((b' - 1)L + a,\ b')$ for every target b'.*

The existing proof is already parametric in the target: the at most L globally close pairs each align fewer than b' challenges, removing them costs at most $(b' - 1)L$, and the weak property applied to the rest gives a contradiction. The case $b' = n + 1$ is the existing statement.

Corollary 6. *With $m = \max(\lfloor \delta n \rfloor + 1, 2)$, strong line decodability with parameters $(a, 2t)$, list size at most t at radius $2t\delta/(2t-1)$, $n < q$ and $(t+1)^2 \leq mt + a$ give $\varepsilon_{\mathrm{mca}}(C, \delta) \leq (mt + a)/q$. The existing bound is $(nt + a)/q$.*

Corollary 7. *Under the hypotheses of [1, Cor. 4.10] as formalized in ArkLib (`frs_mcaError_le`: admissible folding with $s > 4t^2$, $sn < q$, a generator ω of $\mathbb{F}^\times$, $t > 0$), with $\rho = k/(sn)$ and $\delta = 1 - \rho - 2/t$,*

$$\varepsilon_{\mathrm{mca}}(C, \delta) \leq \frac{(\delta n + 2)t + 3t^3}{q},$$

and therefore $\varepsilon_{\mathrm{mca}}(C,\delta) \le \min\bigl(nt+3t^3,\ (\delta n+2)t+3t^3\bigr)/q$. *The second bound is smaller exactly when* $(1-\delta)n > 2$.

ρ	t	δ	n	ratio of bounds	bits
1/2	10	0.30	2^{20}	3.33	1.74
1/2	20	0.40	2^{20}	2.50	1.32
1/2	50	0.46	2^{20}	2.16	1.11
1/4	20	0.65	2^{20}	1.54	0.62
1/8	20	0.775	2^{20}	1.29	0.37

Table 1: Ratio $(nt+3t^3)/((\delta n+2)t+3t^3)$ of the existing and new FRS bounds. The gain is close to $1/\delta$ when $nt \gg t^3$. These are upper bounds on the MCA error, not measured soundness of a protocol.

6 Formalization

All statements are proved in Lean 4 (toolchain v4.35.0) against ArkLib commit `2203b01`. New declarations: `card_le_floor_add_one_of_pencil_mismatch` (Lemma 3), `IsLineDecodable.mcaError_le_of_floor_add_two` (Theorem 4), `strongLineDecodable_boost_of_lambda_le_target` (Lemma 5), `mcaError_le_of_strongLineDecodable_of_floor_add_two` (Corollary 6), `frs_mcaError_le_floor_add_two` and `frs_mcaError_le_min` (Corollary 7). The existing theorems keep their statements; in the final version `IsLineDecodable.mcaError_le` and the original boosting lemma are derived from the same parametrised proofs. `#print axioms` for every new declaration lists only `propext`, `Classical.choice` and `Quot.sound`; ArkLib's full validation script (`validate.sh --axioms`) passes. [Repository link and commit to be added at release.]

7 Related work

The threshold $\ell n+1$ appears in [1, Thm. 3.5] and in [3, Thm. 2.10]. The MCA bound of [2, Thm. 2.13] has the form $(\ell n L+a)/|K|$; we expect the same refinement to apply for $\ell = 1$ but have not checked it against its proof. Jeronimo, Liu and Rajpal [4] use a related agreement count for line correlated agreement. For degree-ℓ curves the counting argument gives $|E| \le \ell(e+1)$, since a mismatch coordinate is a root for at most ℓ challenges; we checked this numerically for small cases but did not formalize it.

Acknowledgements and use of AI tools

The counting lemma follows `exists_common_affine_set` of the Proximity Prize submission by BitWonka. The formalization builds on ArkLib and its contributors. Claude (Anthropic) wrote most of the Lean code and the numerical checks; ChatGPT (OpenAI) reviewed the statements, quantifier structure and literature comparison. The author checked every statement and is responsible for the content; the Lean kernel checks every proof.

References

[1] R. Goyal and V. Guruswami. Optimal proximity gaps for subspace-design codes and (random) Reed–Solomon codes. ECCC TR25-166 (2025, revised 2026); IACR ePrint 2025/2054.

[2] O. Alrabiah, R. Goyal and V. Guruswami. Explicit Reed–Solomon subspace designs: optimal list and line decoding. ECCC TR26-262; arXiv:2610.36215, October 2026.

[3] R. Goyal, V. Guruswami, Y. Sun and M. Wootters. Locality of curve-decoding and improved proximity gaps. arXiv:2607.08516, 2026; APPROX/RANDOM 2026, LIPIcs vol. 392, paper 61.

[4] F. G. Jeronimo, L. Liu and P. Rajpal. Optimal proximity gap for folded Reed–Solomon codes via subspace designs. arXiv:2601.10047, 2026.

[5] Proximity Prize submission `016274d1-9b09-4ce1-add3-4c632dbbc283` (68.15 bits; listed under BitWonka on the leaderboard), `ProximityPrize/SubmissionLower` in `https://github.com/proximity-prize/proximity-prize`, commit `eb82d3b`, Apache-2.0.

[6] ArkLib contributors. ArkLib: a library for formally verified arguments of knowledge. `https://github.com/Verified-zkEVM/ArkLib`.